ServerCats

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Agenda

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3. Proposed Solution and Project Overview
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AUTOMATED LINUX-BASED HOSPITAL SERVER MANAGEMENT SYSTEM

Problem Statement

Manual server management risks data breaches, unauthorized access, and data loss.

A secure, automated system is needed to ensure data protection and reliability.

Objective

A hospital requires a centralized Linux server to manage doctors, nurses, and administrative staff securely.
Without proper management, the system may face:

- Shared user accounts and poor accountability.
- Unauthorized access to sensitive departmental data.
- Missing or unverified data backups.
- Insecure remote access and server configurations.
- Lack of continuous system monitoring.

Solution

- Automates user creation and group management using Linux scripts.
- Implements role-based access control with ACL permissions.
- Schedules disk monitoring, cleanup and daily backups.
- Provides automated recovery and security testing.
- Uses a React dashboard to display server status and PASS/FAIL results.

Project Overview

Orange Cats is a Linux-based hospital administration system that automates server setup, access control, storage management and monitoring.

- **Server Automation**: Configures users, groups, storage, permissions and security.
- **Access Control**: Provides role-based access for doctors, nurses and hospital administrators.
- **Backup & Recovery**: Automates daily backups, disk checks and data restoration.
- **Web Dashboard**: Displays server health and results from 29 automated checks.
- **Verification**: Provides real command outputs and PASS/FAIL status with automatic updates.

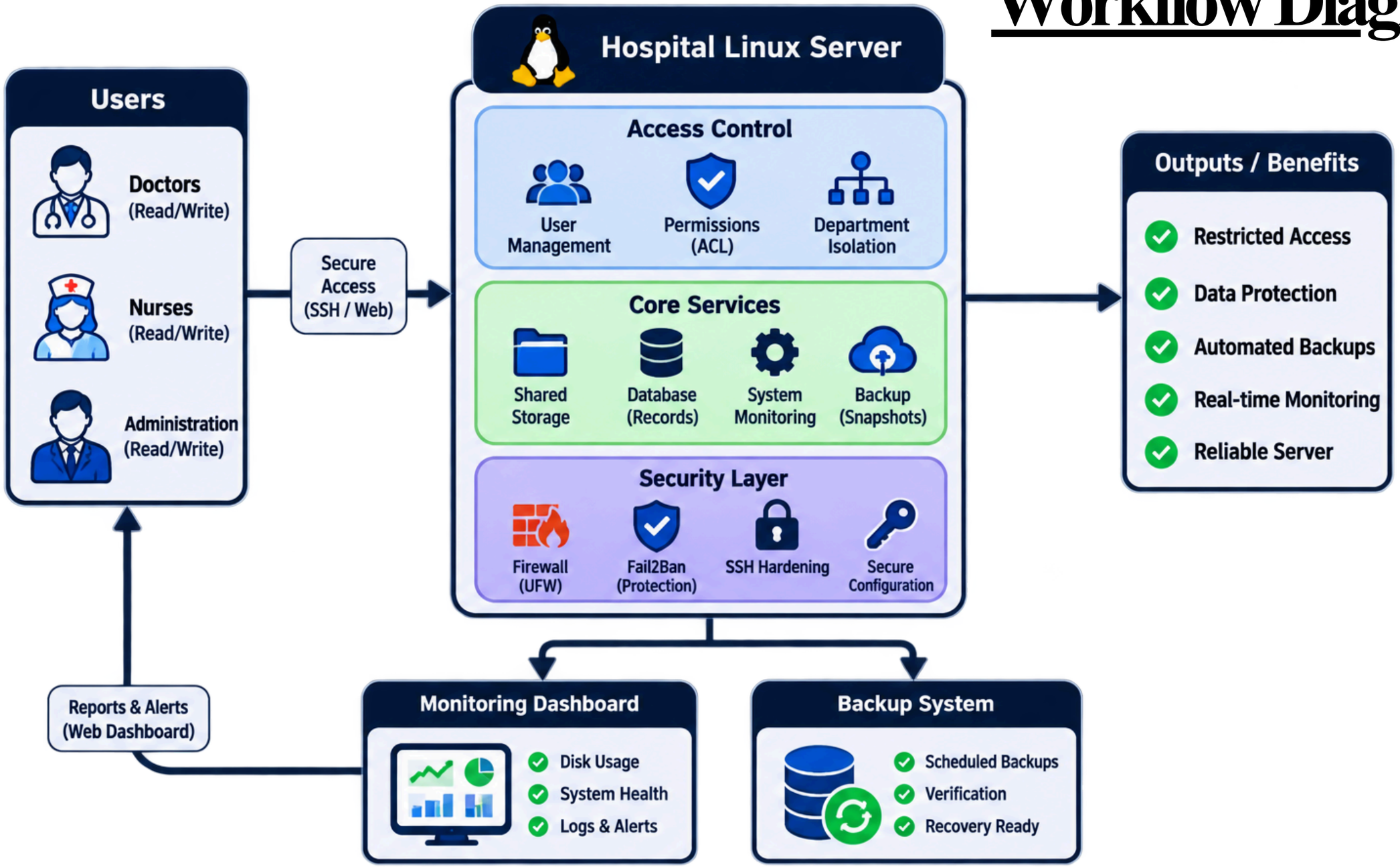
Core Objective: Simplify hospital server administration through secure automation, controlled access and centralized monitoring.

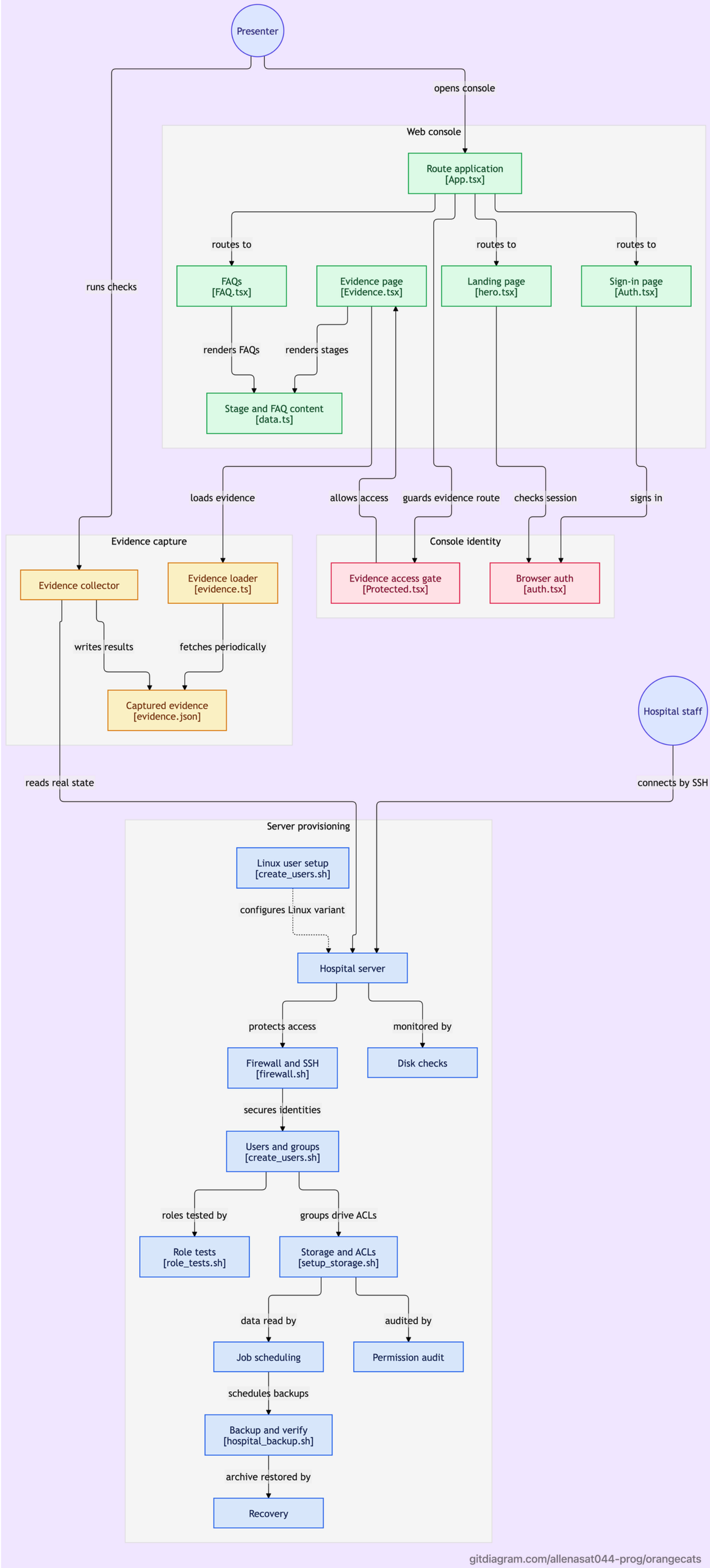
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SYSTEM ARCHITECTURE

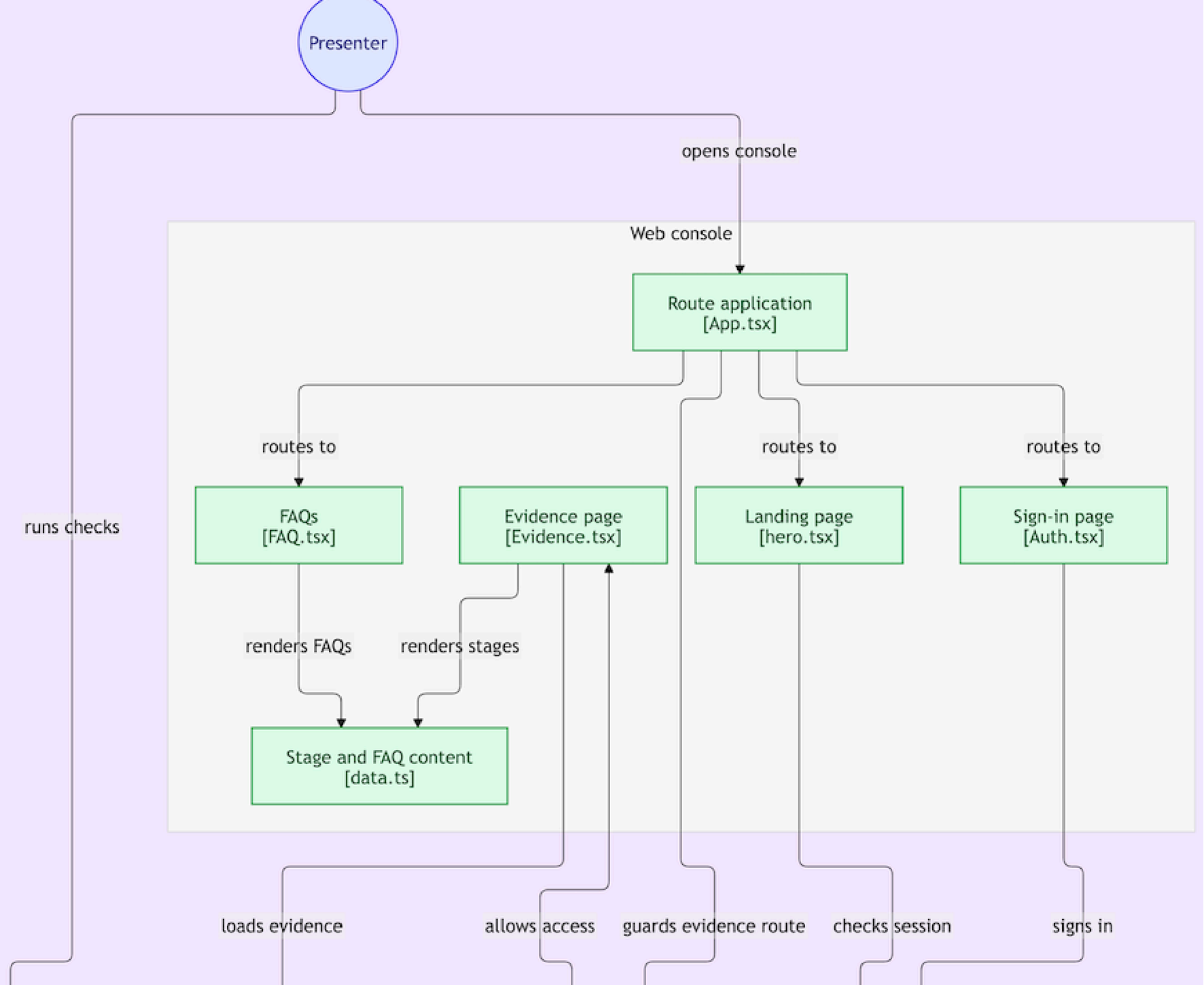


Workflow Diagram

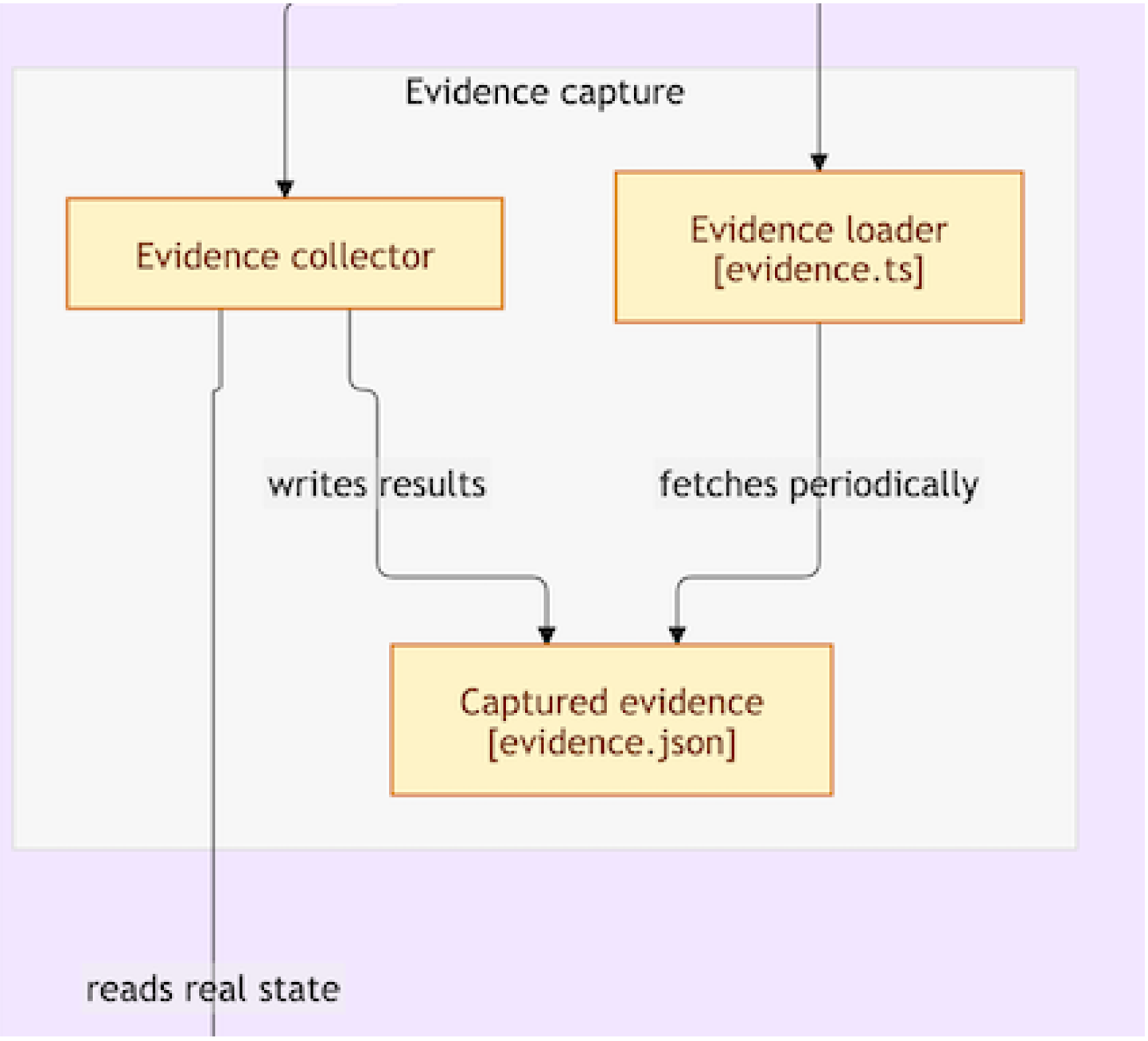




gitdiagram.com/allenasat044-prog/orangecats

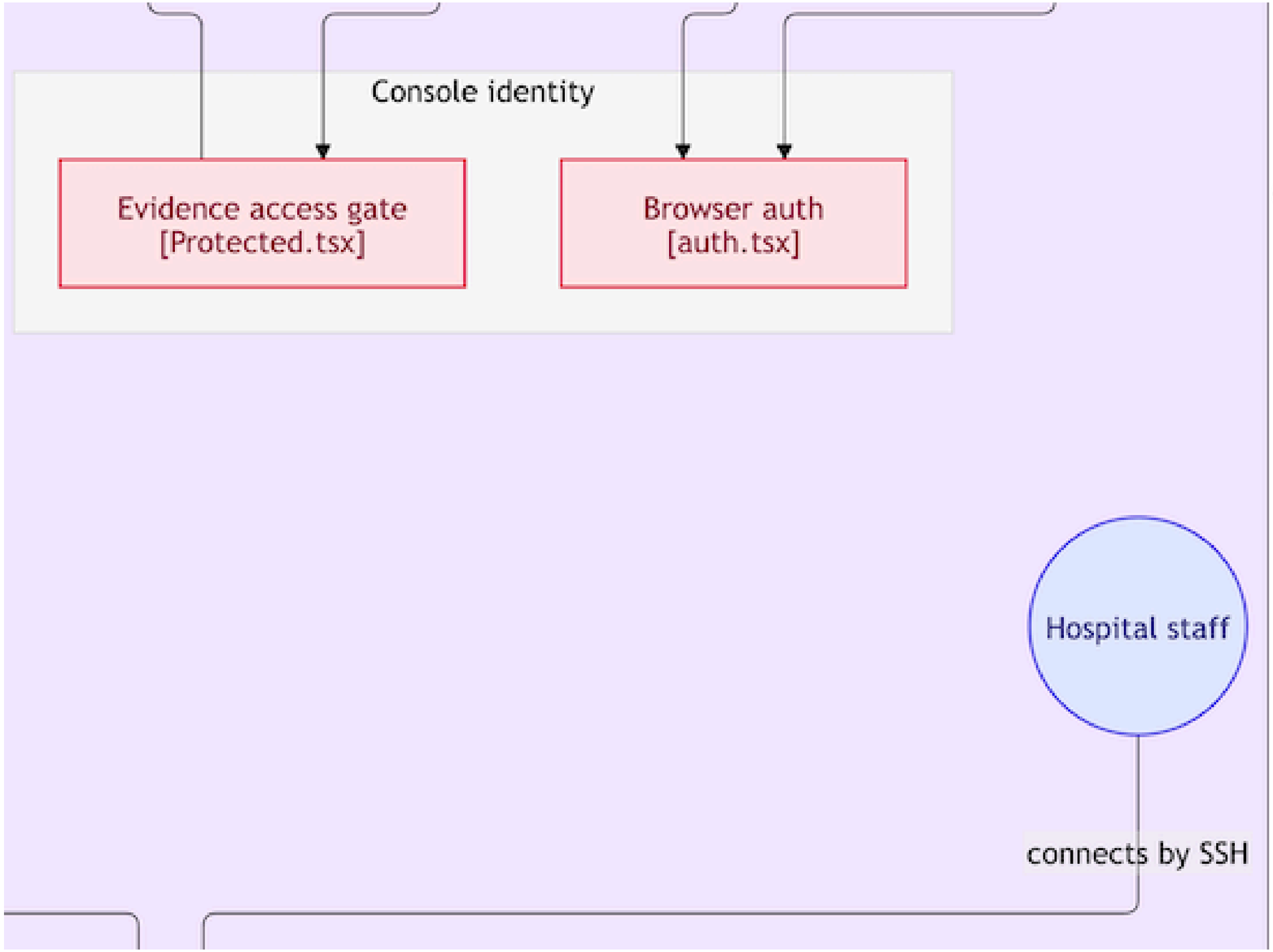

Presenter
opens console
runs checks
Web console
Route application
[App.tsx]
routes to
routes to
routes to
FAQs
[FAQ.tsx]
Evidence page
[Evidence.tsx]
Landing page
[hero.tsx]
Sign-in page
[Auth.tsx]
renders FAQs
renders stages
Stage and FAQ content
[data.ts]
loads evidence
allows access
guards evidence route
checks session
signs in

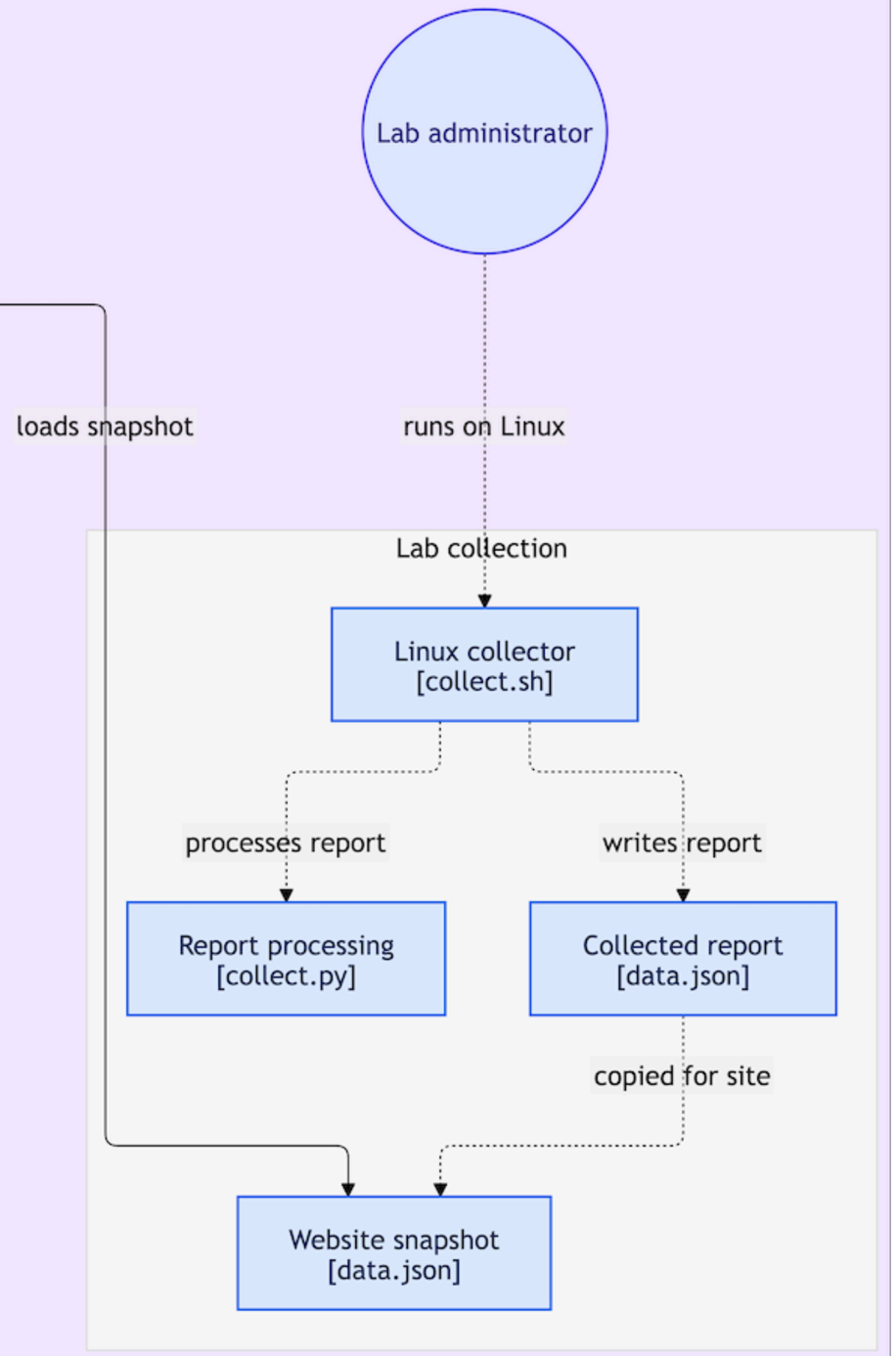
Web Console



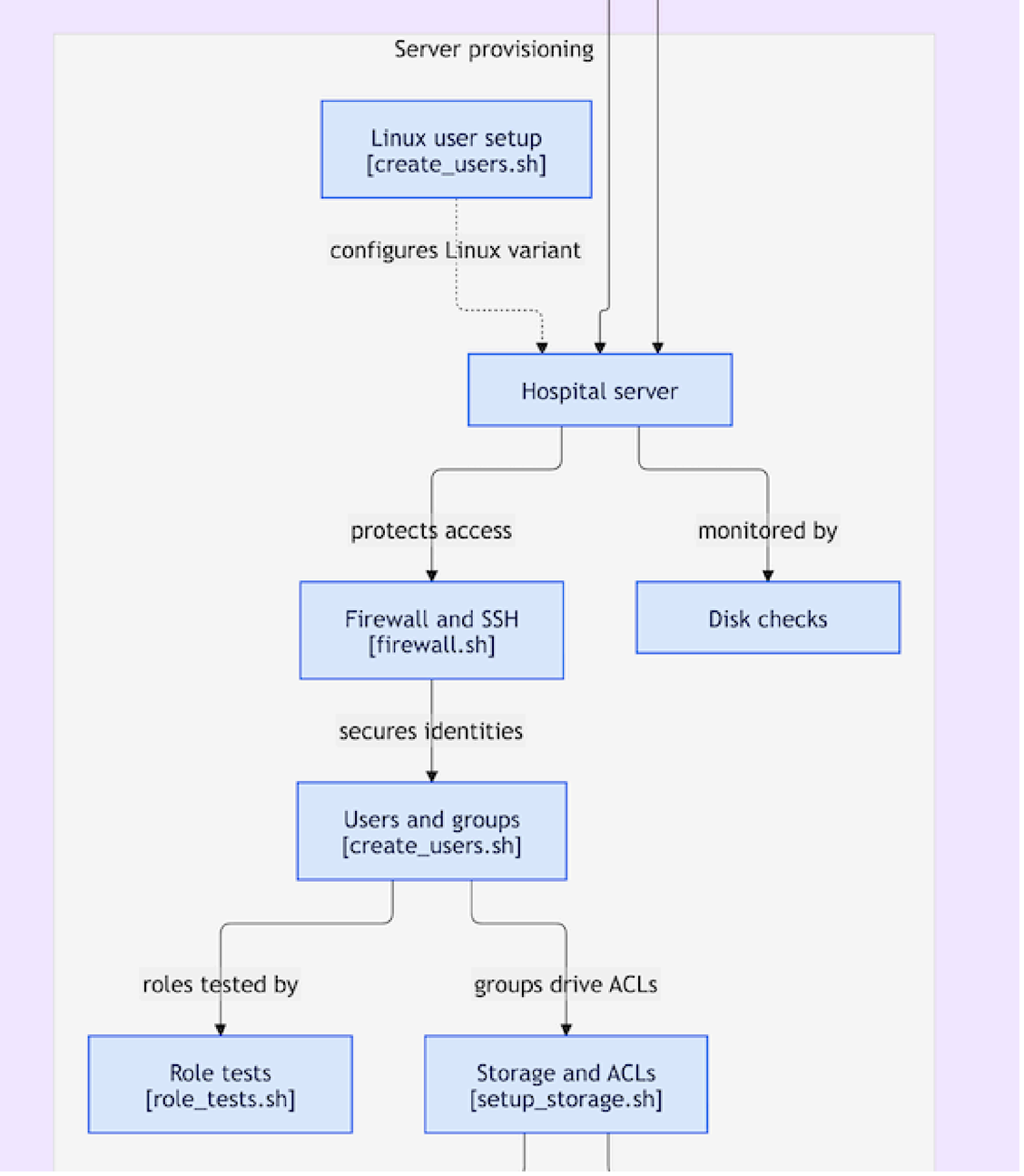
Evidence Capture

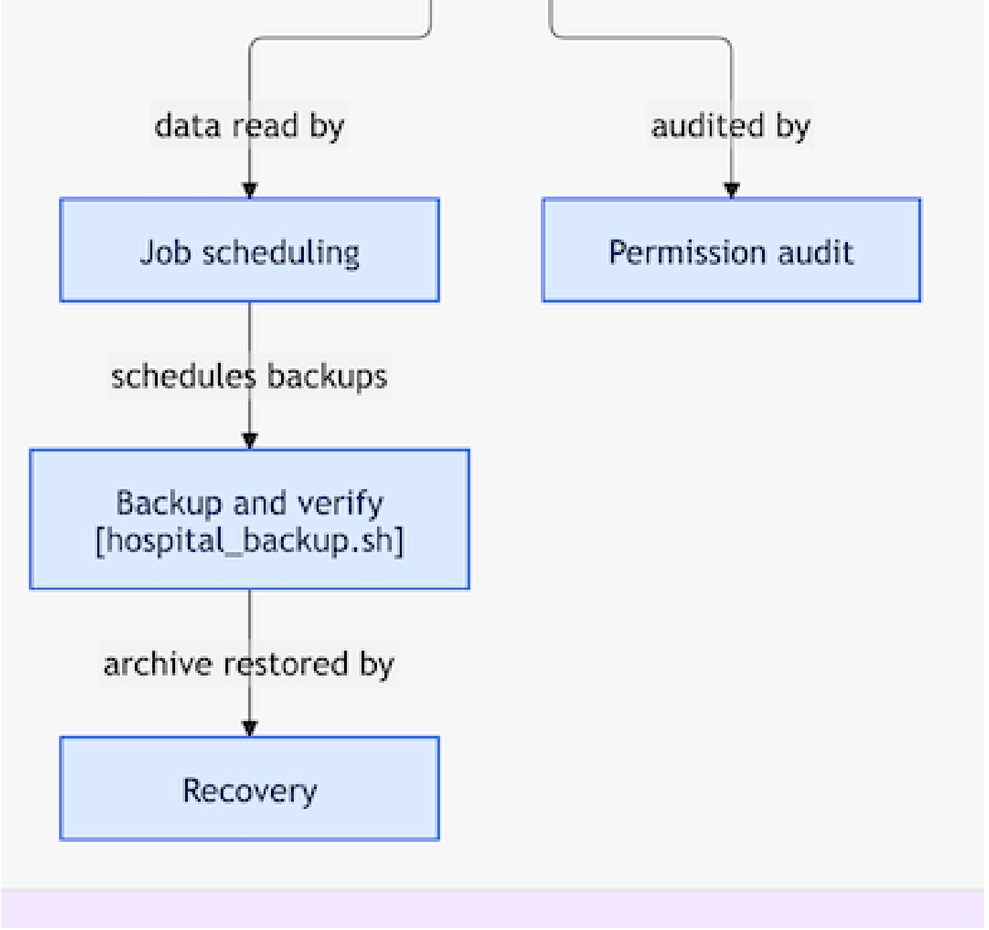
Console Identity & SSH





Lab Collections





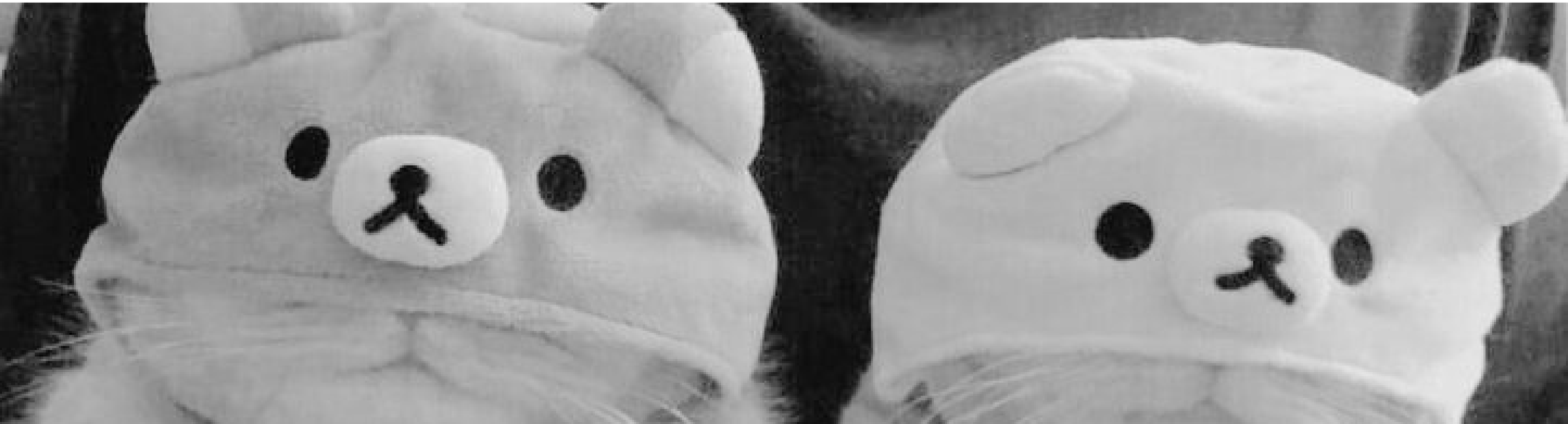
Server Provisioning

Tools, Technologies and Linux Environment Used

Area	macOS (tested)	Linux equivalent	Windows equivalent
OS	macOS (Apple Silicon/Intel)	Ubuntu 22.04 / 24.04	Windows 10/11 Pro
Accounts & groups	`dscl`, `dseditgroup`	`useradd`, `groupadd`, `getent`	`New-LocalUser`, `New-LocalGroup`
Storage	Sparse disk image (APFS), mounted at `/Volumes/hospital`	**LVM** on loop device (`pvcreate`, `vgcreate`, `lvcreate`, ext4)	VHDX via `diskpart` (NTFS)
Folder permissions	`chmod +a` ACLs	`setfacl` / `getfacl`	`icacls`
Scheduling	`launchd`	`cron` (`/etc/cron.d`)	Task Scheduler
Backup integrity	`tar`, `shasum`	`tar --acls`, `sha256sum`	`Compress-Archive`, `Get-FileHash`
Firewall	`pf` (own anchor)	`ufw`	Windows Firewall
Remote access	`sshd_config.d` drop-in	`sshd_config.d` drop-in	OpenSSH Server
Web console	Vite, React, TypeScript, Tailwind CSS, oxlint	same	same

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LINUX COMMANDS & SCRIPTS



LINUX COMMANDS & SCRIPTS

7.2 LVM implementation (Linux / Ubuntu)

```
sudo truncate -s 2G /srv/hospital.img
LOOP=$(sudo losetup --find --show /srv/hospital.img)
sudo pvcreate "$LOOP"
sudo vgcreate hospital_vg "$LOOP"
sudo lvcreate -n data -l 100%FREE hospital_vg
sudo mkfs.ext4 /dev/hospital_vg/data
sudo mkdir -p /hospital
sudo mount /dev/hospital_vg/data /hospital
sudo mkdir -p /hospital/{patient-records,nursing-notes,billing,announcements}
sudo chmod 700 /hospital/{patient-records,nursing-notes,billing,announcements}
```

LINUX COMMANDS & SCRIPTS

7.3 ACL configuration (Linux)

The `-d` entries are default ACLs so new files inherit the rules.

```
# patient-records: doctors rw, nurses read
sudo setfacl -m g:doctors:rwx,g:nurses:r-x /hospital/patient-records
sudo setfacl -d -m g:doctors:rwx,g:nurses:r-x /hospital/patient-records

# nursing-notes: nurses rw, doctors read
sudo setfacl -m g:doctors:r-x,g:nurses:rwx /hospital/nursing-notes
sudo setfacl -d -m g:doctors:r-x,g:nurses:rwx /hospital/nursing-notes

# billing: administration only
sudo setfacl -m g:admin:rwx /hospital/billing
sudo setfacl -d -m g:admin:rwx /hospital/billing

# announcements: everyone reads, administration writes
sudo setfacl -m g:doctors:r-x,g:nurses:r-x,g:admin:rwx /hospital/announcements
sudo setfacl -d -m g:doctors:r-x,g:nurses:r-x,g:admin:rwx /hospital/announcements

getfacl /hospital/patient-records
```

BACKUP

7.4 Backup strategy

Property	Value
Schedule	Daily at **02:00**
Format	Dated `tar.gz` archive (ACLs preserved)
Integrity	**SHA-256** checksum stored and verified right after creation
Retention	Last **7** archives kept, older deleted
Location	Outside the data volume (`/Users/Shared/hospital-backups` on macOS, `/var/backups/hospital` on Linux)

Automation and Shell Scripting

- Backup script (Linux version)
- Disk-space monitor (Linux version)
- Scheduling
- Evidence automation

Security and User/Permission Management

User and permission model

- Three groups, six users, all created as standard accounts. None is an administrator (verified with dseditgroup -o checkmember -m nurse_meera admin → no).
- Each user has a private home folder; shared data lives only in the group-controlled folders.
- Least privilege is enforced with ACLs rather than broad chmod 777-style permissions. Folders are 700 at base, with explicit per-group ACL entries.

Firewall

Linux (ufw):

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow from 192.168.0.0/24 to any port 22 proto tcp
sudo ufw enable
```

macOS (`pf`): `firewall.sh on` blocks inbound traffic except SSH from the local network, uses its own `pf` anchor, and **automatically rolls back if the network check fails**, so you cannot lock yourself out.

```
sudo bash src/scripts/mac/firewall.sh check
sudo bash src/scripts/mac/firewall.sh on
sudo bash src/scripts/mac/firewall.sh status
```

Conclusion and Future Enhancements

Future Enhancements

- Cloud-based off-site backups.
- Multi-factor authentication (MFA).
- Real-time SMS/email alerts.
- Advanced monitoring and analytics.
- Hospital database integration.

Conclusion

- Automated user and department management.
- Implemented secure access and SSH protection.
- Integrated backups and system monitoring.
- Developed a centralized web dashboard.

```
shashank@Shashanks-Laptop hospital-it % sudo bash src/scripts/mac/harden_ssh.sh apply
SSH hardening applied.
Allowed users: shashank dr_asha dr_ravi nurse_meera nurse_john admin_priya admin_kiran
Allowed users added to the macOS SSH access group.
shashank@Shashanks-Laptop hospital-it % sudo bash src/scripts/mac/permission_audit.sh
PASS: ownership is enabled on /Volumes/hospital
PASS: patient-records is drwx------ root:wheel
PASS: patient-records has 2 ACL entries
PASS: nursing-notes is drwx------ root:wheel
PASS: nursing-notes has 2 ACL entries
PASS: billing is drwx------ root:wheel
PASS: billing has 1 ACL entries
PASS: announcements is drwx------ root:wheel
PASS: announcements has 3 ACL entries
PASS: no world-writable items on /Volumes/hospital
PASS: dr_asha is a standard user
PASS: dr_ravi is a standard user
PASS: nurse_meera is a standard user
PASS: nurse_john is a standard user
PASS: admin_priya is a standard user
PASS: admin_kiran is a standard user
PASS: backup folder is drwx------ root:wheel
PASS: SSH root login is disabled
PASS: firewall is enabled with the hospital rules

All checks passed.
```

```
shashank@Shashanks-Laptop hospital-it % sudo bash src/scripts/mac/role_tests.sh
user         patient-records   nursing-notes     billing           announcements
dr_asha      rw ok             r ok              - ok              r ok
dr_ravi      rw ok             r ok              - ok              r ok
nurse_meera  r ok              rw ok             - ok              r ok
nurse_john   r ok              rw ok             - ok              r ok
admin_priya  - ok              - ok              rw ok             rw ok
admin_kiran  - ok              - ok              rw ok             rw ok

All role checks passed.
shashank@Shashanks-Laptop hospital-it % sudo bash src/scripts/mac/role_tests.sh
user         patient-records   nursing-notes     billing           announcements
dr_asha      rw ok             r ok              - ok              r ok
dr_ravi      rw ok             r ok              - ok              r ok
nurse_meera  r ok              rw ok             - ok              r ok
nurse_john   r ok              rw ok             - ok              r ok
admin_priya  - ok              - ok              rw ok             rw ok
admin_kiran  - ok              - ok              rw ok             rw ok

All role checks passed.
```

```
shashank@Shashanks-Laptop hospital-it % sudo bash src/scripts/mac/collect_evidence.sh
Collecting evidence from this Mac. This takes a few seconds.

PASS  s1-hostname  Server hostname
PASS  s1-network  Network settings
PASS  s1-os  Operating system
PASS  s1-updates  System updates
PASS  s1-tools  Administration tools
PASS  s1-rsync  rsync
PASS  s2-groups  Group membership
PASS  s2-ids  User identities
PASS  s2-notadmin  No administrator rights
PASS  s2-homes  Private home folders
PASS  s2-whoami  Running as a sample user
PASS  s3-volume  Storage volume
PASS  s3-acl  ACL rules
PASS  s3-denied  Access is denied where it should be
PASS  s3-allowed  Access is granted where it should be
PASS  s4-jobs  Scheduled jobs
PASS  s4-backup-log  Latest backups
PASS  s4-backup-files  Backup files
PASS  s4-verify  Backup integrity
PASS  s4-disk  Disk usage
PASS  s4-disk-warning  Disk warning works
PASS  s5-firewall  Firewall
PASS  s5-ssh-settings  SSH settings in effect
PASS  s5-ssh-file  SSH hardening file
PASS  s5-remote-login  Remote Login
PASS  s5-access-group  macOS SSH access list
PASS  s5-audit  Permission audit
PASS  s6-roles  Role tests
PASS  s6-restore  Delete and restore

29 of 29 checks passed, 0 failed.
Evidence saved to /Users/shashank/Desktop/hospital-it/public/evidence.json
Open the Evidence page in the console to see it.
```

▼ PASS **Permission audit**

Every folder, ACL, backup folder, SSH setting and firewall rule was checked.

```
$ bash "/Users/shashank/Desktop/hospital-it/src/scripts/mac/permission_audit.sh"

PASS: ownership is enabled on /Volumes/hospital
PASS: patient-records is drwx------ root:wheel
PASS: patient-records has 2 ACL entries
PASS: nursing-notes is drwx------ root:wheel
PASS: nursing-notes has 2 ACL entries
PASS: billing is drwx------ root:wheel
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PASS: announcements is drwx------ root:wheel
PASS: announcements has 3 ACL entries
PASS: no world-writable items on /Volumes/hospital
PASS: dr_asha is a standard user
PASS: dr_ravi is a standard user
PASS: nurse_meera is a standard user
PASS: nurse_john is a standard user
PASS: admin_priya is a standard user
PASS: admin_kiran is a standard user
PASS: backup folder is drwx------ root:wheel
PASS: SSH root login is disabled
PASS: firewall is enabled with the hospital rules
```

Exit code 0

Stage 6: Testing, recovery and demo 2/2

▼ PASS **Role tests**

All six users get exactly the access the matrix gives them on all four folders.

```
$ bash "/Users/shashank/Desktop/hospital-it/src/scripts/mac/role_tests.sh"

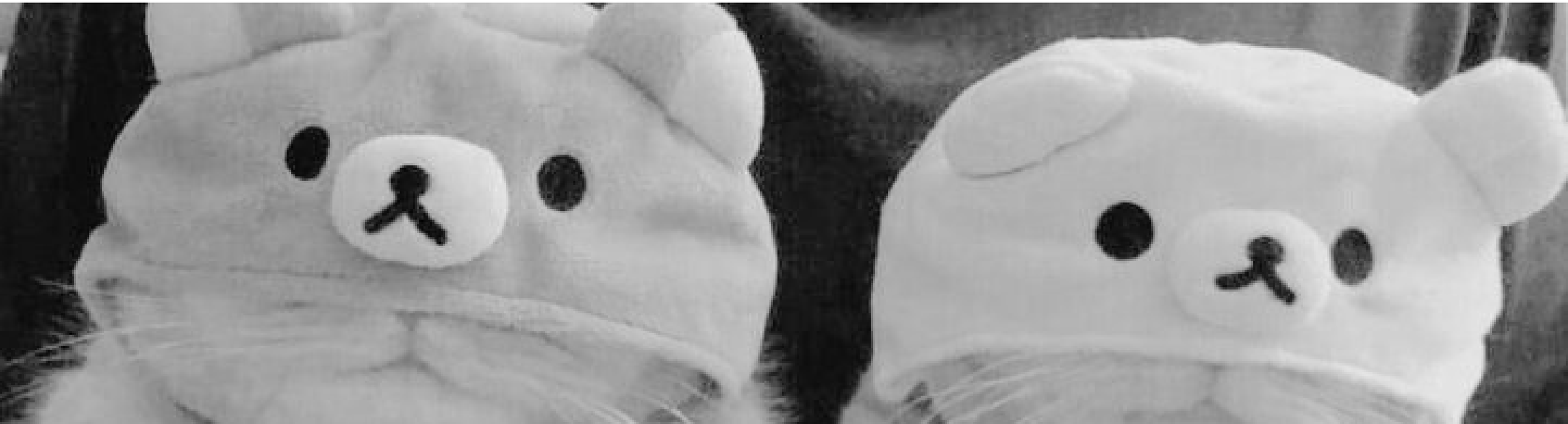
user         patient-records   nursing-notes     billing          announcements
dr_asha      rw ok             r ok              - ok             r ok
dr_ravi      rw ok             r ok              - ok             r ok
nurse_meera  r ok              rw ok             - ok             r ok
nurse_john   r ok              rw ok             - ok             r ok
admin_priya  - ok              - ok              rw ok            rw ok
admin_kiran  - ok              - ok              rw ok            rw ok

All role checks passed.
```

Exit code 0

▼ PASS **Delete and restore**

TESTING AND VALIDATION



Challenges Faced and Solutions

Challenge	Cause	Solution
Standard users could not SSH in; connection closed after the password	macOS keeps its **own SSH access list** that blocked standard users	Found via logs; `harden_ssh.sh` now adds the users to the macOS SSH access group
Hardening status check missed a setting	Upper/lower-case mismatch in the config check	Fixed case-insensitive matching in the script
`admin` group name clash	macOS `admin` = full administrator rights	Named the group `hosp_admin`
No LVM on macOS	LVM is Linux-only	Sparse disk image as a resizable, unmountable volume; full LVM commands documented for Linux
Volume missing after restart	Disk images are unmounted on reboot	Re-run `setup_storage.sh` (idempotent)
Risk of firewall locking out the admin	Wrong `pf` rule could block the network	Added `check` mode and automatic rollback if the network test fails
Internet stops after the firewall step	`pf` rules blocked something	`firewall.sh off`, then adjust
Missed backups when the laptop sleeps	cron-style timing does not fit sleeping machines	Used `launchd`, which runs missed jobs on wake
Web page cannot run commands	Browser sandbox	Evidence pipeline: script → `evidence.json` → console polling
`systemsetup` refused to change Remote Login	Terminal lacked Full Disk Access	Enable Remote Login in System Settings → General → Sharing

THANK YOU